

UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Extended Relational Algebra

Basi di Dati

Bachelor's Degree in Computer Engineering
Academic Year 2026/2027



DIPARTIMENTO
DI INGEGNERIA
DELL'INFORMAZIONE

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Outline

- Extended relational algebra
- Some examples

Introduction

- In traditional relational algebra, operations evaluate one tuple at time, and **independently** of all the other tuples in the relation.

- However, when we query a database, we may need information that is not explicitly stored as the value of any attribute.

- **Selections, projections, joins** may not be enough.

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Introduction

Pizzeria

Pizza	Prezzo	Quantità
Margherita	8	10
Capricciosa	11	20
Funghi	9	2
Marinara	6	13

- How can we know the **total income** for each type of pizza?

Introduction

Pizzeria

Pizza	Prezzo	Quantità
Margherita	8	10
Capricciosa	11	20
Funghi	9	2
Marinara	6	13

- How can we know the **total income** for each type of pizza?
- We need to be able to compute operations between the attributes of our relations
- We need to compute **transformations on data.**

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Introduction

Pizzeria

ID	Sede	Pizza	Prezzo	Quantità
1	Vicenza	Margherita	8	10
2	Padova	Capricciosa	11	20
3	Padova	Margherita	9	2
4	Vicenza	Marinara	6	13

- How can we compute the **income** for each location?
- We need to be able to compute operations between the attributes of our relations
- We need to compute **aggregation functions on grouped data.**

Generalized projection

Given a relation $R(X)$ and $F_1, F_2, F_3, \ldots, F_n$ functions over the attributes of the relation, the **generalized projection** is the relation

$$\pi_{F_1,F_2,F_3,\ldots,F_n}(R(X)) = \{(F_1(t), F_2(t), F_3(t), \ldots, F_n(t)) \mid t \in R\}$$

- The generalized projection extends the standard projection by allowing transformations or computations on attributes.
- Instead of just selecting columns, it can apply functions or expressions to generate new values in the result set.
- The projection will be characterized by:
  - the attributes of R declared in the projection (constant values);
  - one or more functions on the attributes of R.

Generalized projection

Pizzeria

Pizza	Prezzo	Quantità
Margherita	8	10
Capricciosa	10	20
Funghi	9	2
Marinara	6	13

- We want to compute the income related to each type of pizza.

$$\pi_{pizza,prezzo*quantita}(Pizzeria)$$

Generalized projection

Pizzeria

Pizza	Prezzo	Quantità
Margherita	8	10
Capricciosa	10	20
Funghi	9	2
Marinara	6	13

$$\pi_{pizza,prezzo*quantita}(Pizzeria)$$

Costi Pizza (Derivata)

Pizza	
Margherita	80
Capricciosa	200
Funghi	18
Marinara	78

Generalized projection

Costi Pizza (Derivata)

Pizza	
Margherita	80
Capricciosa	200
Funghi	18
Marinara	78

$$\rho_{totale \leftarrow prezzo * quantita}(\pi_{pizza, prezzo * quantita}(Pizzeria))$$

Costi Pizza (Derivata)

Pizza	Totale
Margherita	80
Capricciosa	200
Funghi	18
Marinara	78

Aggregation functions

- A type of request that cannot be expressed in the basic relational algebra requires to specify mathematical **aggregate functions** on collections of values from the database
- In the context of relational algebra, aggregation functions are operations applied to sets of tuples in order to compute **summary information.**
- These functions operate not on individual tuples — as basic relational algebra does — but on **entire groups of tuples**
- Typical aggregation functions include **SUM**, **AVG**, **MIN**, **MAX**, and **COUNT**.

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Aggregation functions

- Aggregation functions:
  - **SUM:** sum values contained in a column of a table
  - **MAX:** find the max value within the values of an attribute
  - **MIN:** find the min value within the values of an attribute
  - **AVERAGE:** compute the average of the values of an attribute
  - **COUNT:** count the number of tuples in a relation. This operation is a bit different as it counts **tuples** and not the values of an attribute.

Aggregation Operation

Given a relation $R(X)$ and $f_1(A_i), \ldots, f_n(A_j)$ aggregation functions over the attributes of R with $A_i, A_j \in X$, the **aggregation operation** is:

$$\mathscr{F}_{f_1(A_i),\ldots,f_n(A_j)}(R(X))$$

- $\mathscr{F}$ is the aggregation operator
- The result is a **new relation** with n attributes (as many as the aggregation functions), with exactly **one tuple** which is the result of the operations.
- degree = n, cardinality = 1

Exercise

Find the total income and the max total.

$$INCOME \leftarrow \rho_{totale \leftarrow prezzo*quantita}(\pi_{pizza,prezzo*quantita}(Pizzeria))$$

INCOME

Pizza	Totale
Margherita	80
Capricciosa	200
Funghi	18
Marinara	78

$$\mathcal{F}_{SUM(totale),MAX(totale)}(INCOME)$$

SUM(totale)	MAX(totale)
376	200

Exercise

Find the total income and the max total.

$$\rho_{totale_income,max_income \leftarrow SUM(totale),MAX(totale)}(\mathcal{F}_{SUM(totale),MAX(totale)}(INCOME))$$

Totale income	Max income
376	200

Grouping

- Up to now, aggregation has been applied to the **whole** set of tuples in a relation.

- However, we may instead want to compute aggregations **separately on subsets of tuples**, where each subset is determined by the values of one or more attributes.

- **Grouping** is an operation that divides a relation into groups based on the values of one or more attributes and then applies an aggregation function to each group.

Grouping Operation

Given a relation $R(X)$ and $f_1(A_i), \ldots, f_n(A_j)$ aggregation functions over the attributes of R with $A_i, A_j \in X$, given m grouping attributes, the **grouping operation** is:

$$(A_1,A_1,\ldots,A_m)\mathscr{F}_{f_1(A_i),\ldots,f_n(A_j)}(R(X))$$

- $\mathscr{F}$ is the aggregation operator
- The attributes before the $\mathscr{F}$ are the grouping attributes, that are used to group the tuples in the relation $R(X)$
- degree = $n + m$, cardinality = # groups

Exercise

Restaurant

Name	Date	price_menu	Seats
Ristorante_1	10/12/24	80	100
Ristorante_2	10/12/24	90	200
Ristorante_2	11/12/24	70	300
Ristorante_1	11/12/24	100	150
Ristorante_3	11/12/24	60	70

We collect the total of menus ordered in three restaurants in two days. We want to get, for each restaurant, the total and the average incomes.

Exercise

Restaurant

Name	Date	price_menu	Seats
Ristorante_1	10/12/24	80	100
Ristorante_2	10/12/24	90	200
Ristorante_2	11/12/24	70	300
Ristorante_1	11/12/24	100	150
Ristorante_3	11/12/24	60	70

$$\rho_{income \leftarrow price_menu * seats}((\pi_{name,date,price_menu * seats}(Restaurant)))$$

Name	Date	Income
Ristorante_1	10/12/24	800
Ristorante_2	10/12/24	1800
Ristorante_2	11/12/24	2100
Ristorante_1	11/12/24	1500
Ristorante_3	11/12/24	4200

Exercise

$$INCOME \leftarrow \rho_{income \leftarrow price_menu * seats}((\pi_{name,date,price_menu * seats}(Restaurant)))$$

Income

Name	Date	Income
Ristorante_1	10/12/24	800
Ristorante_2	10/12/24	1800
Ristorante_2	11/12/24	2100
Ristorante_1	11/12/24	1500
Ristorante_3	11/12/24	4200

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Exercise

$$INCOME \leftarrow \rho_{income \leftarrow price_menu * seats}((\pi_{name,date,price_menu * seats}(Restaurant)))$$

INCOME

Name	Date	Income
Ristorante_1	10/12/24	800
Ristorante_2	10/12/24	1800
Ristorante_2	11/12/24	2100
Ristorante_1	11/12/24	1500
Ristorante_3	11/12/24	4200

$$(name)\mathcal{F}_{SUM(income),AVERAGE(income)}(INCOME)$$

Name	SUM(Income)	AVERAGE(Income)
Ristorante_1	2300	1150
Ristorante_2	3900	1950
Ristorante_3	4200	4200

Exercise

$$INCOME \leftarrow \rho_{income \leftarrow price_menu * seats}((\pi_{name,date,price_menu * seats}(Restaurant)))$$

INCOME

Name	Date	Income
Ristorante_1	10/12/24	800
Ristorante_2	10/12/24	1800
Ristorante_2	11/12/24	2100
Ristorante_1	11/12/24	1500
Ristorante_3	11/12/24	4200

$$\rho_{total,average \leftarrow SUM(income),AVERAGE(income)}({}_{(name)}\mathcal{F}_{SUM(income),AVERAGE(income)}(INCOME))$$

Name	Total	Average
Ristorante_1	2300	1150
Ristorante_2	3900	1950
Ristorante_3	4200	4200

NULL Values and Aggregation Operations

EXAM

Student	Date	Mark	Course
123456	10/12/24	20	Analysis
123456	10/12/24	25	Databases
384752	11/12/24	NULL	Analysis
384752	11/12/24	18	Physics
3841345	11/12/24	NULL	Physics

The aggregation operations such as AVERAGE, MAX, MIN, SUM does not consider NULL values, and the result is computed considering exclusively the tuples whose values for the attributes in aggregation are **not null.**

NULL Values and Aggregation Operations

EXAM

Student	Date	Mark	Course
123456	10/12/24	20	Analysis
123456	10/12/24	25	Databases
384752	11/12/24	NULL	Analysis
384752	11/12/24	18	Physics
3841345	11/12/24	NULL	Physics

The aggregation operations such as AVERAGE, MAX, MIN, SUM does not consider NULL values, and the result is computed considering exclusively the tuples whose values for the attributes in aggregation are **not null.**

NULL Values and Aggregation Operations

Badge	Name	Surname	Car
123456	Mario	Rossi	Volvo
384762	Matteo	Verdi	NULL
382320	Marco	Verdi	NULL
596830	John	Rossi	NULL
192092	Mary	Bianchi	Fiat
384135	Isabel	Bianchi	Fiat

$$|_{(badge,surname)}\mathcal{F}_{COUNT(*)}(Worker)| = 6$$

Badge	Surname	COUNT(*)
123456	Rossi	1
384762	Verdi	1
382320	Verdi	1
596830	Rossi	1
192092	Bianchi	1
384135	Bianchi	1

NULL Values and Aggregation Operations

Badge	Name	Surname	Car
123456	Mario	Rossi	Volvo
384762	Matteo	Verdi	NULL
382320	Marco	Verdi	NULL
596830	John	Rossi	NULL
192092	Mary	Bianchi	Fiat
384135	Isabel	Bianchi	Fiat

$$|_{(surname,Car)}\mathcal{F}_{COUNT(*)}(Worker)| = 4$$

Surname	Car	COUNT(*)
Rossi	Volvo	1
Verdi	NULL	2
Rossi	NULL	1
Bianchi	Fiat	2

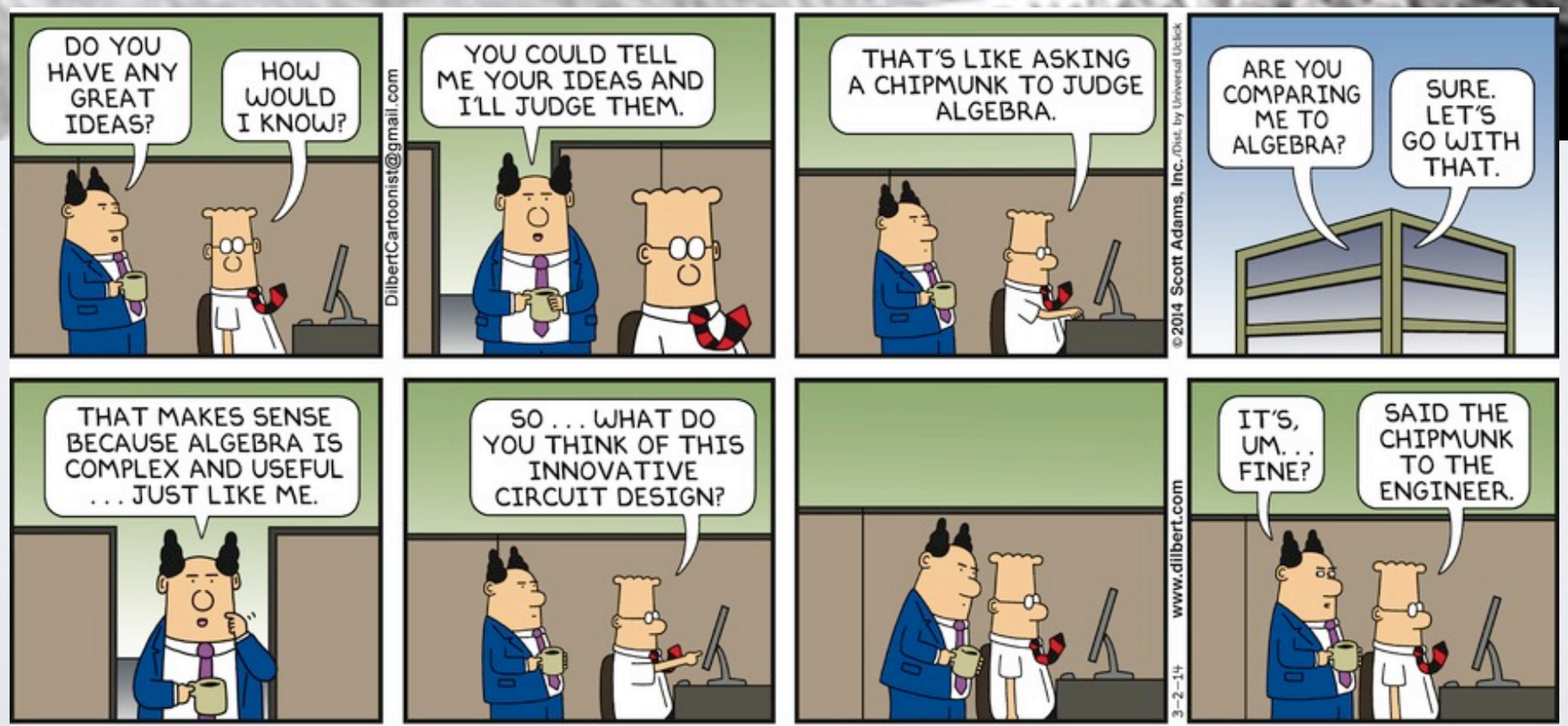
NULL Values and Grouping Operations

Badge	Name	Surname	Car
123456	Mario	Rossi	Volvo
384762	Matteo	Verdi	NULL
382320	Marco	Verdi	NULL
596830	John	Rossi	NULL
192092	Mary	Bianchi	Fiat
384135	Isabel	Bianchi	Fiat

$$|_{(Car)}\mathcal{F}_{COUNT(*)}(Worker)| = 3$$

Car	COUNT(*)
Volvo	1
Fiat	2
NULL	3

Questions?


DO YOU HAVE ANY GREAT IDEAS?
HOW WOULD I KNOW?
DilbertCartoonist@gmail.com
YOU COULD TELL ME YOUR IDEAS AND I'LL JUDGE THEM.
THAT'S LIKE ASKING A CHIPMUNK TO JUDGE ALGEBRA.
ARE YOU COMPARING ME TO ALGEBRA?
SURE. LET'S GO WITH THAT.
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THAT MAKES SENSE BECAUSE ALGEBRA IS COMPLEX AND USEFUL . . . JUST LIKE ME.
SO . . . WHAT DO YOU THINK OF THIS INNOVATIVE CIRCUIT DESIGN?
IT'S, UM. . . FINE?
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